Introduction to Computer system

Binary Addition:-

$0+0=0$

$0+1=1$

$1+0=1$

$1+1=10$ → sum

carry

	carry	sum
$1+0+0=$	0	1
$1+1+0=$	1	0
$1+0+1=$	1	0
$1+1+1=$	1	1

Ex

①

$$\begin{array}{r} 1\;1\;\;\; \\ 1\;1 \\ 1^{+} \\ \hline 1\;0\;0 \end{array} \equiv \begin{array}{r} 3 \\ 1^{+} \\ \hline 4 \end{array}$$

②

$$\begin{array}{r} 1\;1\;\;\; \\ 1\;1 \\ 1\;1^{+} \\ \hline 1\;1\;0 \end{array} \equiv \begin{array}{r} 3 \\ 3^{+} \\ \hline 6 \end{array}$$

```
    1   1
③  1 1 0 1
     1 0 1 +
  ----------
  1 0 1 0

   1       1
④  1 0 0 0 1
   1 1 1 0 1 +
  ------------
  1 0 1 1 1 0

  1   1 1 1
⑤  1 0 1 1 1
  1 1 0 0 0 1 +
  -------------
  1 0 0 1 0 0 0
```

Binary subtraction

$0 - 0 = 0$

$1 - 1 = 0$

$1 - 0 = 1$

$0 - 1 = 1$ borrow of 1

Ex ①

$$\begin{array}{r} 1\ 1 \\ 0\ 1^{-} \\ \hline 1\ 0 \end{array} \equiv \begin{array}{r} 3 \\ 1^{-} \\ \hline 2 \end{array}$$

②

$$\begin{array}{r} \overset{0}{\cancel{1}}\ \overset{2}{0}\ 1 \\ 0\ 1\ 1^{-} \\ \hline 0\ 1\ 0 \end{array}$$

③

$$\begin{array}{r} 1\ \overset{0}{\cancel{1}}\ \overset{2}{0}\ 1\ 1 \\ 1\ 0\ 1\ 1\ 0^{-} \\ \hline 0\ 0\ 1\ 0\ 1 \end{array} \equiv \begin{array}{r} 27 \\ 22^{-} \\ \hline 5 \end{array}$$

Binary Multiplication

$0 * 0 = 0$

$0 * 1 = 0$

$1 * 0 = 0$

$1 * 1 = 1$

steps

1. Multiply
2. Add

Ex

①

$$\begin{array}{r} 11 \\ 11\,* \\ \hline {\scriptstyle 1\ 1}\quad\ \ \\ 011 \\ 110\,+ \\ \hline 1001 \end{array} \equiv \begin{array}{r} 3 \\ 3\,* \\ \hline 9 \end{array}$$

②

$$\begin{array}{rcccccl}
 & & & 1 & 1 & 1 & \\
 & & & 1 & 0 & 1 & * \\
\hline
 & 1 & 1 & & & & \\
1 & 0 & 0 & 1 & 1 & 1 & \\
0 & 0 & 0 & 0 & 0 & & + \\
1 & 1 & 1 & 0 & 0 & & \\
\hline
1 & 0 & 0 & 0 & 1 & 1 &
\end{array}$$

$$\begin{array}{r}
7 \\
5 * \\
\hline
35
\end{array}$$

Binary Divison

① Division
② Multiply
③ Subtraction

Ex 110/11

①
```
     010
11 ) 110  -
     11↓
     000
```

Compare divisor with dividend
① 11>1 ?
② 11>11 ?
③ 11>000 ?

②
```
     00111
11 ) 10101  -
     11
     100  -
      11
     011  -
      11
        0
```

Ex
```
     071
2 ) 142
    0
    14  -
    14
    02  -
     2
     0
```

Ex basic concept
```
     016
8 ) 128
    0
    12  -
     8
    48  -
    48
     0
```

③

0011

100) 1100

100

100 -

100

0

④

00101

101) $1\overset{0}{1}\overset{2}{0}01$

101 ↓

00101

101 -

0

⑤

00011

1000) 11000

1000 ↓

1000

1000 -

0